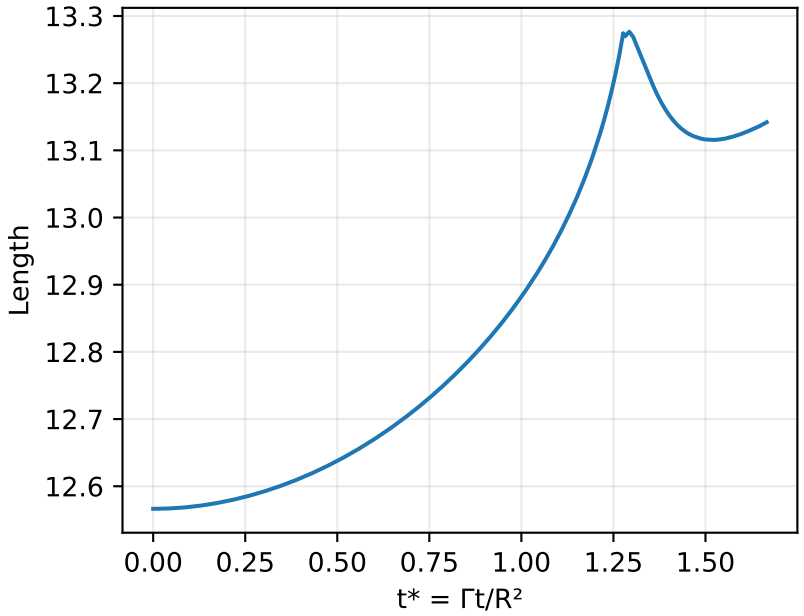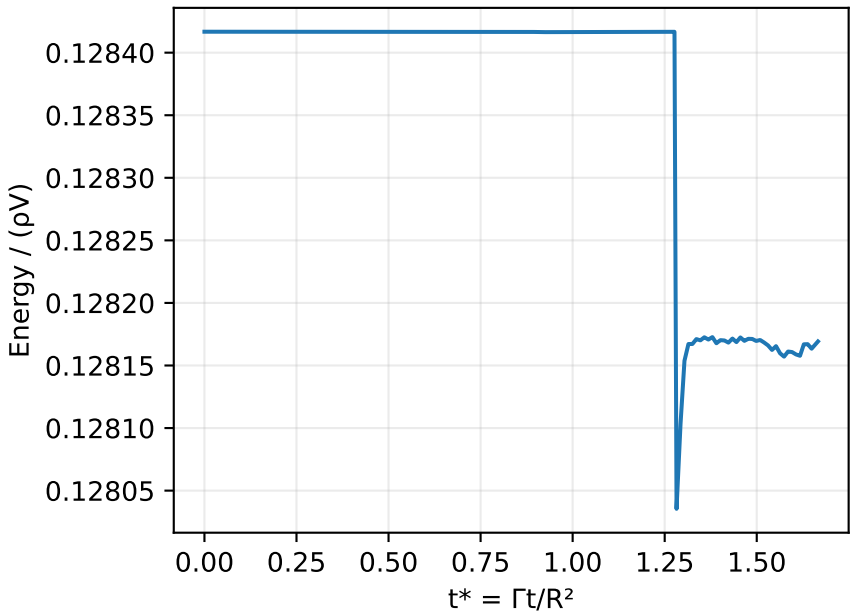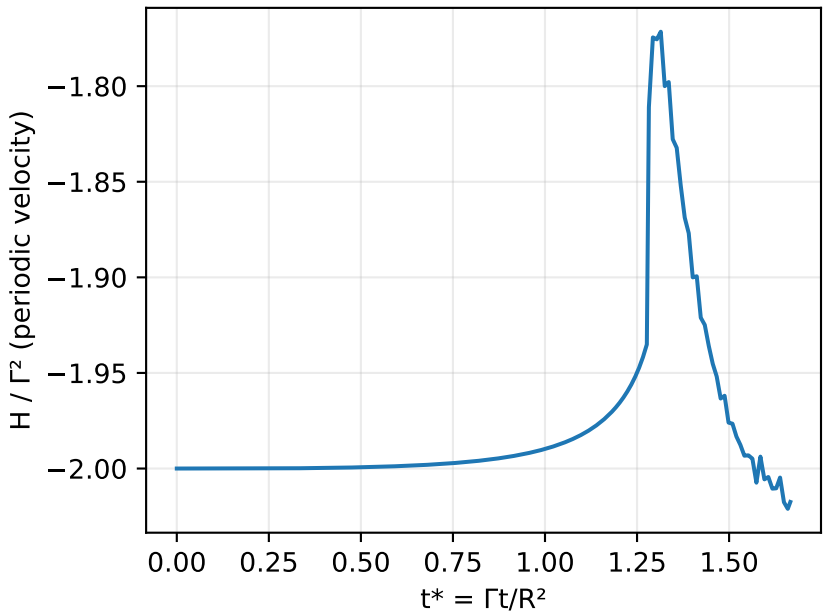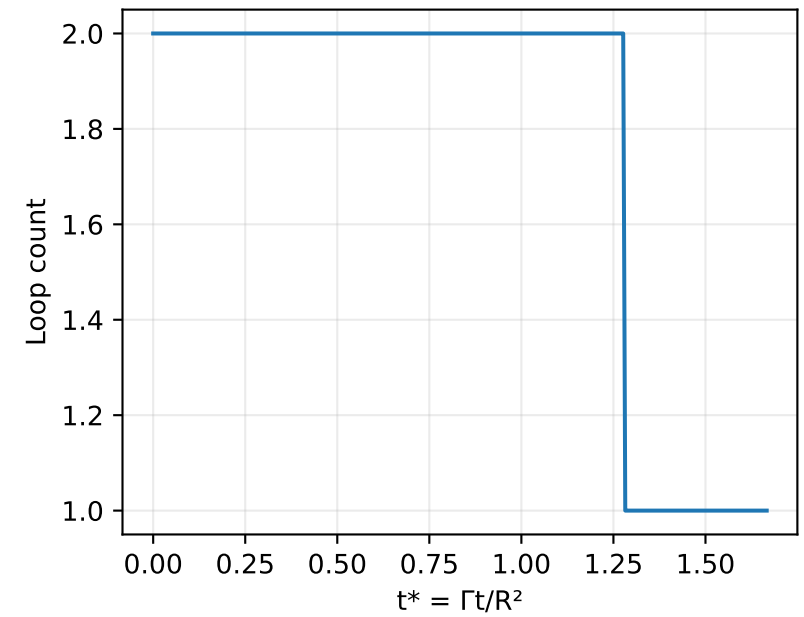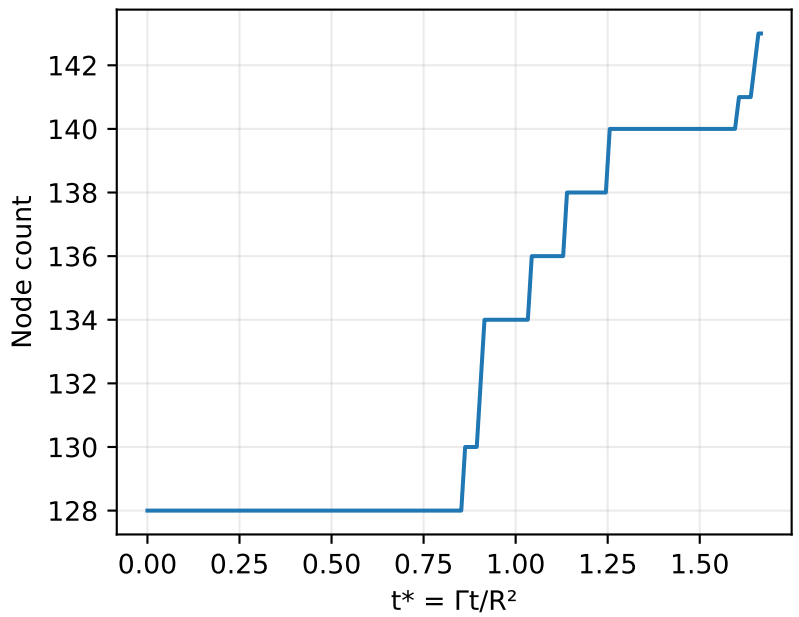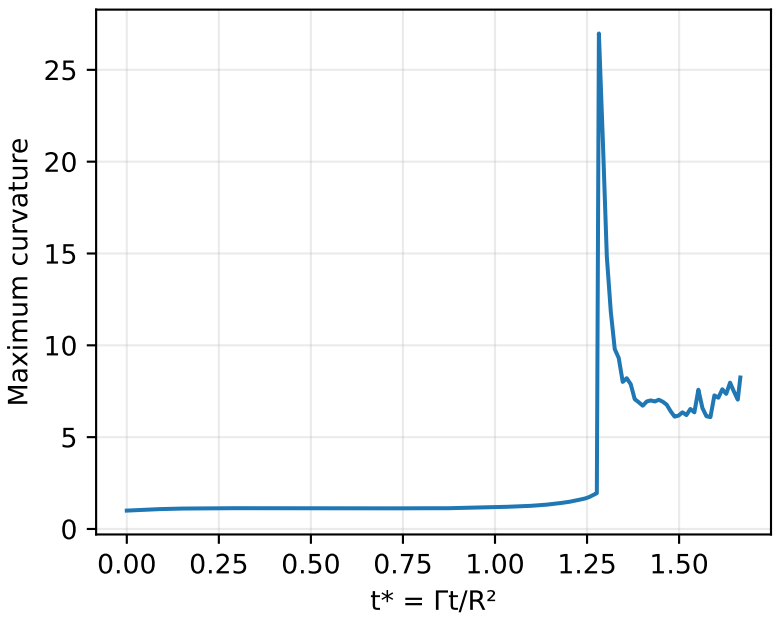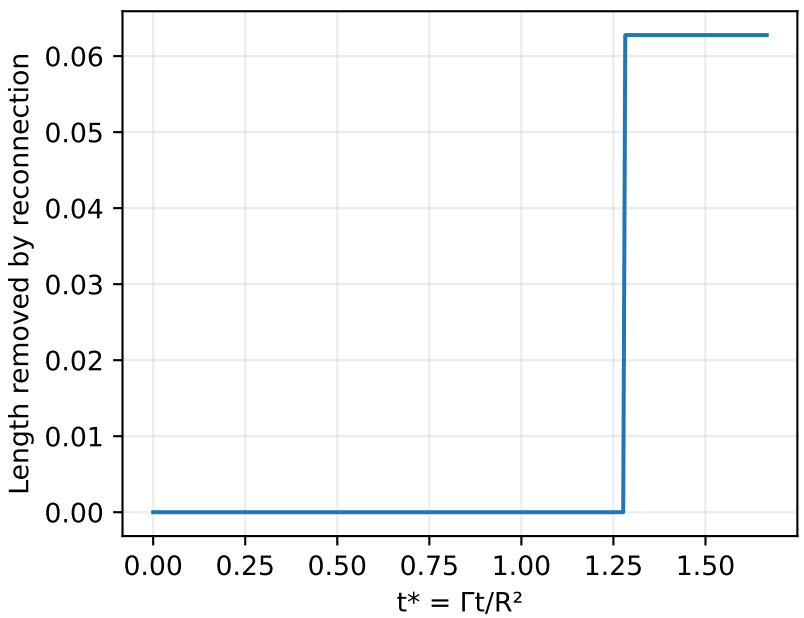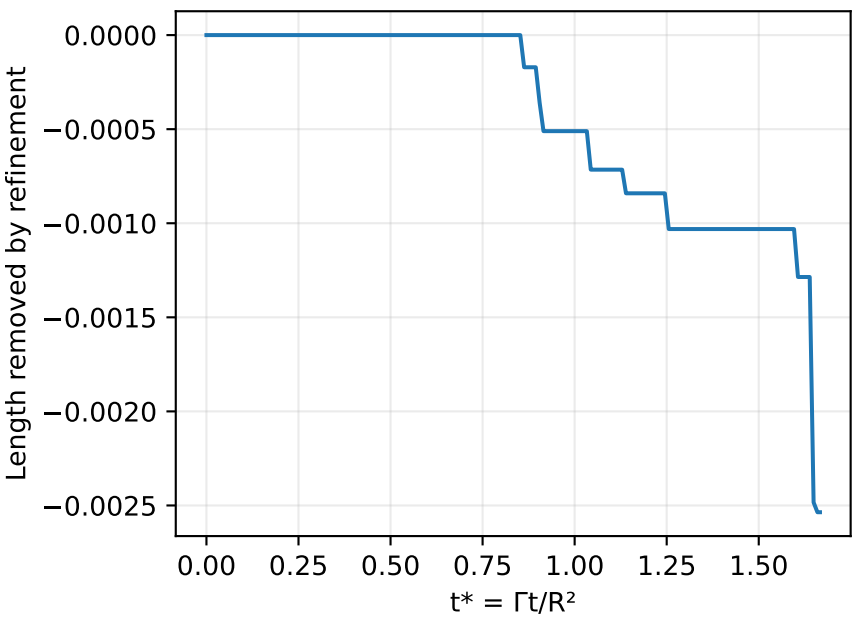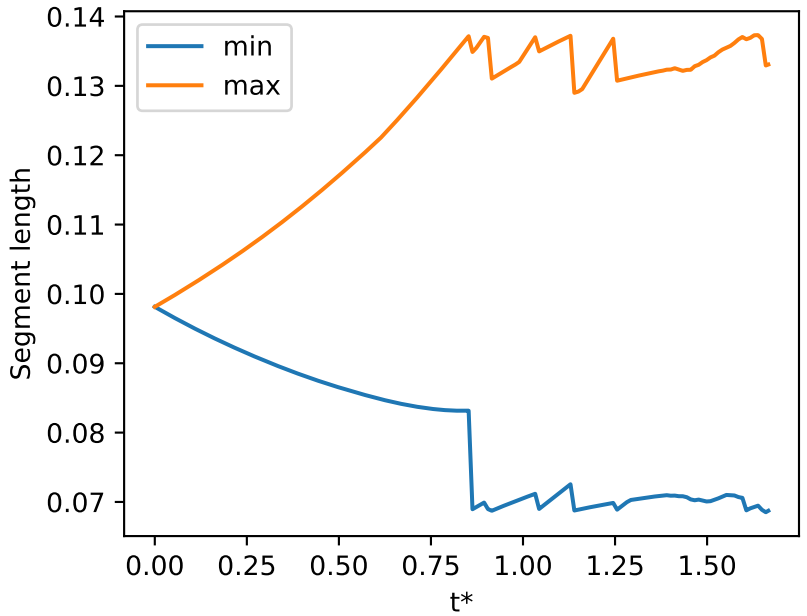hopf_halfdt · a=0.001 · no mutual friction